

Fentanyl

(Sublimaze, Duragesic)

- I. **Classification** – Opioid Analgesics
- II. **Action(s)**
 - Binds to opiate receptors in the CNS
 - Alters the perception of and response to painful stimuli while producing generalized CNS depression
- III. **Indication(s)**
 - Severe pain
 - Post-Intubation Sedation
- IV. **Contraindications**
 - Known hypersensitivity
- V. **Precautions**
 - Severe renal, hepatic, or pulmonary disease
 - History of substance abuse (if alternative is available, otherwise must be given with expressed consent from patient, parent, or guardian, and only in the setting of condition that would be painful)
 - Elderly or debilitated patients (dosage reduction suggested)
- VI. **Drug interactions**
 - MAO inhibitors or linezolid – may result in serotonin syndrome
 - Sedative/Hypnotics, clomiprazine, barbituates, TCAs, and antihistamines – may increase CNS depression
- VII. **Adult Dosing and Administration**
 - 1 mcg/kg IV/IO/IM/IN (max initial dose 100mcg, max total dose 300 mcg)
 - Max initial dose for chest pain is 50mcg
 - Consider 0.5 mcg/kg (max initial dose 25 mcg) in elderly
 - For pain: may repeat x 2 q 5 min
 - For post-intubation sedation
 - May repeat 1mcg/kg IV/IO every 10 minutes PRN
 - OR
 - Initial Fentanyl bolus plus Fentanyl 1-2 mcg/kg/hr (max 200mcg/hr) continuous infusion titrated every 30-60 minutes to max 200mcg/hr for pain control
 - Monitor respiratory and ventilatory status utilizing continuous SpO2 and ETCO2
 - Call On-Call Medical Control Physician for additional doses
- VIII. **Pediatric Dosing and Administration**
 - 1 mcg/kg IV/IO/IM/IN (max initial dose 50mcg, max total 3mcg/kg or 300mcg)
 - May repeat x 2 after 5 minutes
 - For Post Intubation Sedation:
 - Fentanyl 1mcg/kg IV/IO (max 200mcg) every 10 min titrated for pain
 - OR
 - Initial Fentanyl bolus plus Fentanyl 1-2 mcg/kg/hr (max 200mcg/hr) continuous infusion titrated every 30-60 minutes to max 200mcg/hr for pain control
 - Monitor respiratory and ventilatory status utilizing continuous SpO2 and ETCO2
- IX. **Onset**
 - Immediate
- X. **Duration**
 - 30-60 minutes

XI. Possible Adverse Effects

- Allergic Reaction/anaphylaxis
- Respiratory depression/laryngospasm/chest wall rigidity/respiratory arrest
- Bradycardia/hypotension
- Altered Mental Status/Seizures

XII. Other Notes

- Administer slowly! Rapid administration may lead to increased respiratory depression, respiratory muscle rigidity, hypotension, and circulatory change
- Assess LOC, pulse, and respirations before and periodically after administration
- Monitor ETCO₂ after administration